

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Mixed evidence with one decisive negative for this HIGH-priority dimension. INJONGO was designed with cultural grounding as an explicit principle [Q13, Q30, Q37], and dataset inspection confirms Ethiopian-specific entities in the Amharic split: Telebirr as PAYMENT_COMPANY [D4, D11], Ethiopian banks Amara Bank and CBE [D2, D3], Ethiopian cities Dire Dawa, Bahir Dar, Hawassa [D5, D12], and Ethiopian calendar months Hamle and Meskerem [D5, D6] — these resolve key gaps from web search positively. However, the deployment-critical requirement of Amharic-English code-switching is not represented: zero of 19 Amharic samples contain Latin-script tokens or English insertions (D24), and code-switching was not an explicit design criterion in the paper. The deployment elicitation marks code-switching as 'common and increasing' particularly for technical vocabulary (token, top-up, balance, meter), and a benchmark monolingual in Ethiopic script will overestimate performance on naturalistic deployment input. Additionally, regional dialectal variation (Oromo/Tigrinya substrate) is not addressed, and Amharic is acknowledged as systematically lower-resource than Swahili [Q88, Q89].

ID	Evidence
Q13	"However, these efforts face two key challenges: (1) the translationese effect, which makes utterances sound less natural in the target languages (Vanmassenhove et al., 2021; Bizzoni et al., 2020), and (2) the creation of utterances that are less culturally relevant." (p.1)
Q30	"In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)
Q37	"To simplify the process while making the dataset cultural, we provide each annotator with sample sentences from the CLINC dataset (Larson et al., 2019a) with a specified intent type, say "transfer"." (p.3)
Q88	"The performance of some African languages is often higher than others, this is probably connected to the amount of monolingual data available on the web." (p.7)
Q89	"For example, Swahili (swa) with over 1 billion monolingual data (Kudugunta et al., 2023) has 80.6 accuracy point that is comparable performance to English performance (81.1) with Llama 3.3 70B, while other languages have much lower performance." (p.7)
D2	"የአማራ ባንክ አካውንቱ ስንት አለው" — BANK_NAME: አማራ ባንክ — "How much is in my Amara Bank account?" — Ethiopian bank (Amara Bank) named as BANK_NAME slot
D3	"ንግድ ባንክ ላይ የቀረጽ ገንዘብ አለ" — BANK_NAME: ንግድ ባንክ — "Is there money remaining at Commercial Bank?" — CBE (Commercial Bank of Ethiopia) named
D4	"በቴሌብር የብድር አካውንቱ ላይ አሁን ያለብኝ ቀሪ ሂሳብ ምን ያህል ነው።" — PAYMENT_COMPANY: ቴሌብር — "What is the remaining balance on my Telebirr loan account?" — Telebirr named as PAYMENT_COMPANY
D5	"ከድረዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድረዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
D6	"ከሃምሌ 4 እስከ 6 የቤት ኪራይ ማግኘት እችላለሁ?" — DATE: ከሃምሌ 4 እስከ 6 — "Can I get a home car rental from Hamle 4 to 6?" — Ethiopian calendar month (Hamle) used for DATE slot
D11	raw: "ከቴሌ ብር አካውንቱ ገንዘብ ለማን ለማን እንደላኩ ዝርዝሩን አሳየኝ?" text: "Show me history of my telebirr transfers?" — PAYMENT_COMPANY: telebirr — Telebirr confirmed in Amharic source utterance as code-switched reference; PAYMENT_COMPANY slot correctly labeled
D12	raw: "ለቀጣይ የጉብኝት ጉዞ በሃዋሳ እና በአርባ ምንጭ መካከል እንድመርጥ..." text: "help me decide between Hawassa and Arbaminch..." — CITY_OR_PROVINCE: Hawassa, Arbaminch — Ethiopian regional cities (Hawassa, Arba Minch) correctly identified in travel context
D24	All 19 Amharic examples use Ethiopic script without any romanized characters or Latin-script code-switching — No code-switched (Amharic + English intrasentential) utterances observed in 19 Amharic train examples

Strengths

- Explicit cultural-grounding methodology, with native-speaker generation rather than translation [Q37, Q40, Q41]
- Telebirr confirmed as PAYMENT_COMPANY value in Amharic data (D4) — a high-priority deployment entity
- Ethiopian banks (Amara Bank, CBE) and Ethiopian regional cities (Dire Dawa, Bahir Dar, Hawassa) appear as slot values [D2, D3, D5, D12]
- Ethiopian calendar months (Hamle, Meskerem) appear as DATE values [D5, D6], reflecting authentic Amharic temporal expression
- Formal polite Amharic register present in samples (D29), matching deployment norms for formal service interaction

ID	Evidence
Q37	"To simplify the process while making the dataset cultural, we provide each annotator with sample sentences from the CLINC dataset (Larson et al., 2019a) with a specified intent type, say "transfer". " (p.3)
Q40	"A Xhosa annotator was asked to generate another utterance belonging to the same intent type but capturing the South African context where the language is spoken." (p.3)
Q41	"Thus, the annotator used the R200 as "money" with currency Rand (R), and more familiar South African banks such as "FNB" and "Absa" for "bank name" slot." (p.3)
D2	"የአማራ ባንክ አካውንቲ ስንት አለው" — BANK_NAME: አማራ ባንክ — "How much is in my Amara Bank account?" — Ethiopian bank (Amara Bank) named as BANK_NAME slot
D3	"ንግድ ባንክ ላይ የቀረጽ ገንዘብ አለ" — BANK_NAME: ንግድ ባንክ — "Is there money remaining at Commercial Bank?" — CBE (Commercial Bank of Ethiopia) named
D4	"በቴሌብር የብድር አካውንቲ ላይ አሁን ያለብኝ ቀሪ ሂሳብ ምን ያህል ነው።" — PAYMENT_COMPANY: ቴሌብር — "What is the remaining balance on my Telebirr loan account?" — Telebirr named as PAYMENT_COMPANY
D5	"ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ።" — CITY_OR_PROVINCE: ድሬዳዋ, ባህርዳር — "Book me a flight from Dire Dawa to Bahir Dar for the fifth of Meskerem." — Ethiopian cities and Ethiopian calendar month
D6	"ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ።" — DATE: ከሃምሌ 4 እስከ 6 — "Can I get a home car rental from Hamle 4 to 6?" — Ethiopian calendar month (Hamle) used for DATE slot
D12	raw: "ለቀጣይ የጉብኝት ጉዞዬ በሃዋሳ እና በአርባ ምንጭ መካከል እንድመርጥ..." text: "help me decide between Hawassa and Arbaminch..." — CITY_OR_PROVINCE: Hawassa, Arbaminch — Ethiopian regional cities (Hawassa, Arba Minch) correctly identified in travel context
D29	"የእኔ የወጭ አካውንት ቀሪ ሂሳብ ስንት እንደሆነ ሊነግሩኝ ይችላሉ።" — ACCOUNT_TYPE: የወጭ አካውንት — "Can you tell me the remaining balance of my expense account?" — formal polite register (ሊነግሩኝ ይችላሉ)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — deployment requires Ethiopian-specific cultural/geographic knowledge (Ethiopian payment platforms, banks, kebele/sub-city addresses, Ethiopian calendar) and dialectal robustness (Amharic-English code-switching, possible Oromo/Tigrinya substrate).

IC-2: Partially — Ethiopian financial entities (Telebirr D4, Amara Bank D2, CBE D3) and geography [D5, D12] are present, satisfying baseline cultural grounding [Q30]. However, code-switching is absent (D24) and regional dialectal variation is undocumented (gap_id 6 NOT FOUND).

IC-3: Some Western residue persists (e.g., 'taco bell' in Hausa D22), though for the Amharic split the sampled instances appear consistently Ethiopia-grounded. Cross-lingual training data drawn from 'Combined INJONGO' [Q72, Q128] could expose models to non-Ethiopian entities.

IC-4: INSUFFICIENT DOCUMENTATION — paper documents pan-African native-speaker recruitment via Kenya-based logistics [Q44]; whether Amharic annotators were Ethiopia-based or had urban Addis Ababa familiarity is not specified. Would need annotator location/demographic breakdown.

IC-5: Major content issue: zero observed code-switching across 19 Amharic examples (D24) despite deployment confirming high code-switching frequency for English domain vocabulary. This represents a content distribution mismatch likely to harm external validity. Secondary issue: Amharic web-data resource asymmetry vs. higher-resource African languages may compound performance gaps [Q88, Q89].

ID	Evidence
Q30	"In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)
Q41	"Thus, the annotator used the R200 as "money" with currency Rand (R), and more familiar South African banks such as "FNB" and "Absa" for "bank name" slot." (p.3)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q72	"The dataset of SFT was obtained by aggregating all the training samples of the 17 languages in INJONGO i.e. "Combined INJONGO", with randomly sampled prompts from a pool of 5." (p.6)
Q88	"The performance of some African languages is often higher than others, this is probably connected to the amount of monolingual data available on the web." (p.7)
Q89	"For example, Swahili (swa) with over 1 billion monolingual data (Kudugunta et al., 2023) has 80.6 accuracy point that is comparable performance to English performance (81.1) with Llama 3.3 70B, while other languages have much lower performance." (p.7)
Q128	"Training data is constructed from the combined train splits of INJONGO dataset across all 17 languages, with prompts randomly sampled from a pool of 5 predefined templates." (p.14)
D2	"የአማራ ባንክ አካውንቲ ስንት አለው" — BANK_NAME: አማራ ባንክ — "How much is in my Amara Bank account?" — Ethiopian bank (Amara Bank) named as BANK_NAME slot
D3	"ንግድ ባንክ ላይ የቀረጃ ገንዘብ አለ" — BANK_NAME: ንግድ ባንክ — "Is there money remaining at Commercial Bank?" — CBE (Commercial Bank of Ethiopia) named
D4	"በቴሌብር የብድር አካውንቲ ላይ አሁን ያለብኝ ቀሪ ሂሳብ ምን ያህል ነው።" — PAYMENT_COMPANY: ቴሌብር — "What is the remaining balance on my Telebirr loan account?" — Telebirr named as PAYMENT_COMPANY
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D12	raw: "ለቀጣይ የጉብኝት ጉዞዬ በሃዋሳ እና በአርባ ምንጭ መካከል እንድመርጥ..." text: "help me decide between Hawassa and Arbaminch..." — CITY_OR_PROVINCE: Hawassa, Arbaminch — Ethiopian regional cities (Hawassa, Arba Minch) correctly identified in travel context
D22	"Za ku iya tabbatar min ina da ajiyar tebur a taco bell da karfe 7 na yamma" — RESTAURANT_NAME: taco bell — Hausa: "Can you confirm I have a table reservation at Taco Bell at 7 PM?" — Western fast food chain in culturally-specific dataset
D24	All 19 Amharic examples use Ethiopic script without any romanized characters or Latin-script code-switching — No code-switched (Amharic + English intrasentential) utterances observed in 19 Amharic train examples
WEB-5	Telebirr 54.84M users, dominant Ethiopian utility payment channel — confirms deployment relevance of Telebirr coverage (https://techafricanews.com/2025/07/25/ethio-telecom-achieves-83-2m-subscribers-and-73-revenue-surge-as-lead-strategy-concludes/)
WEB-9	INJONGO paper — gap_id 3 NOT FOUND on Amharic annotator domain expertise; gap_id 7 NOT FOUND on Amharic-specific Kappa (https://arxiv.org/abs/2502.09814)
WEB-12	Code-switching NLP literature — no Amharic-English Ethiopian utility code-switching benchmark exists (https://github.com/gentaiscool/code-switching-papers)

Information Gaps

- Frequency of Telebirr/CBE Birr appearances at full-corpus scale (sample shows present but not density)
- Regional dialectal variation in Amharic split (gap_id 6 NOT FOUND) — Addis Ababa vs Dire Dawa/Hawassa/Mekelle representation
- Whether any utility-specific Amharic content beyond bill_balance exists in unsampled portions of the 2,240 train examples
- Code-switching density across full Amharic split — sample of 19 may understate

Requires Expert Verification

- Whether sampled Amharic register and lexical choices reflect EEU/AAWSA customer service interactions specifically
 - Whether informal Addis Ababa neighborhood naming (Bole, Piassa, Megenagna) appears in PLACE_NAME slots in unsampled data
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Remediation

Gap 1: Zero observed Amharic-English code-switched utterances in sampled INJONGO Amharic, despite code-switching being a confirmed high-priority deployment pattern especially for technical vocabulary (token, top-up, balance, meter).

Recommendation: Augment training data with code-switched Amharic-English variants, ideally elicited from younger urban Telegram users; ensure technical English domain vocabulary appears in mixed-script form; evaluate a held-out code-switched test split separately from monolingual.